



CatalystScan

Fleet Monitor Dashboard

Administrator User Guide

Version 2.1 | April 2026

Bostontech India Pvt Ltd

Professional Academic Hardware Solutions

CONFIDENTIAL — For authorized administrators only

1.	Getting Started — Login & Navigation	3
2.	Dashboard Overview — Fleet Statistics	3
3.	Device Management — View & Manage Devices	4
4.	Organization Management	4
5.	License & Scan Credit Management	5
6.	Building .exe Packages for Clients	5
7.	E-Waste Compliance Reporting	6
8.	AMC / Warranty Tracker	6
9.	Reports & Data Export	7
10.	Lab Utilization Analysis	7
11.	Multi-Organization SI Dashboard	8
12.	Alert & Notification Configuration	8
13.	Network Scanner (Feature-Gated)	9
14.	Admin Credentials & Security	9
15.	Troubleshooting & FAQs	10

1.1 Admin Login

The CatalystScan dashboard requires admin authentication. Navigate to:

<https://catalystscan.bostontechindia.in/dashboard/>

Default Admin Credentials:

Role: Super Administrator

Username: admin

Password: CatalystAdmin@2026

Role: Administrator

Username: bostontech

Password: Boston@2026

■ **SECURITY NOTE**

Change default passwords in production by setting ADMIN_PASS env variable.

Sessions are stored in browser sessionStorage and cleared on tab close.

2. Dashboard Overview

The Overview page shows real-time fleet statistics:

- Total Devices — All registered scanner clients across organizations
- Online / Offline — Real-time status based on heartbeat (5-min interval)
- Scans Used — Total scans consumed across the fleet
- Active Alerts — Tamper/theft alerts requiring attention
- Fleet Health — Grade distribution (A/B/C/D/F) with auto-refresh every 30s

3.1 Device List

Navigate to Devices to see all registered scanner clients:

- Hostname, Fingerprint, Organization, Last Seen, Scan Credits
- Click any device row to view detailed hardware specs & scan results
- Add scan credits: Enter fingerprint + count in the Licenses page

Devices auto-register when a scanner .exe first connects to the server.

3.2 DB Explorer

Direct access to MySQL database tables for advanced queries:

- Select table: devices, organizations, scan_results, heartbeats, etc.
- View raw data with JSON expansion for scan_data, health_data columns

4. Organization Management

Navigate to Organizations to manage customer accounts:

- Create Organization — Set org_id, name, contact email, phone, address
- View linked devices and aggregate scan statistics per org
- Feature flags: Enable/disable network scanning per org

■ TIP

Each .exe package is bound to an org_id via org_config.json.
Build custom .exe packages per client from the Build .exe page.

5.1 Scan Credits

Each device starts with a default scan credit allocation (typically 2 scans).

To add more credits:

- Licenses page → Enter device fingerprint + credit count → Add
- Reports → Bulk License Manager → Paste CSV of fingerprint,count
- API: POST /api/dashboard/bulk-license/org/{org_id} for org-wide add

6. Building .exe Packages

Create org-specific scanner executables for distribution:

- Navigate to Build .exe page
- Enter: Organization Name, Org ID, Telemetry URL, Default Scan Limit
- Click Build — downloads a ZIP with the configured CatalystScan.exe

■ DISTRIBUTION

The .exe is self-contained — no Python installation needed on client PCs.

Each .exe auto-registers with the dashboard on first scan.

Telemetry URL should point to your GCP VM: <http://35.244.19.7:8090>

7.1 E-Waste Report

Navigate to E-Waste to see automated disposal categorization:

- ■ Keep — Systems < 5 years old with Grade A/B/C
- ■ Review — Systems 5-8 years old OR Grade D
- ■ E-Waste — Systems > 8 years old OR Grade F

Uses BIOS date to calculate system age. Compliant with MoEFCC Rule 16.

8. AMC / Warranty Tracker

Track Annual Maintenance Contracts and warranty expiry dates:

- Navigate to AMC Tracker → Click "Add AMC Record"
- Fields: Device Fingerprint, Vendor, Contract Type, Start/End Dates, Cost
- Auto-status: ■ Active, ■ Expiring Soon (<30 days), ■ Expired

■ CONTRACT TYPES

warranty — OEM warranty from manufacturer

amc — Annual Maintenance Contract with third-party vendor

camc — Comprehensive AMC (includes parts replacement)

9.1 Export Reports Hub

Navigate to Reports to access all export tools:

- ■ AICTE/UGC Asset Register — CSV with 15 required columns
- ■ E-Waste Report — Links to E-Waste compliance view
- ■ Software Audit — Cross-device software inventory for NAAC/NBA
- ■ Bulk License Manager — CSV upload for mass credit allocation

10. Lab Utilization Analysis

Navigate to Lab Utilization to see PC usage patterns:

- Based on heartbeat frequency — scanner sends pings every 5 minutes
- Categories: ■ High (>60%), ■ Medium (20-60%), ■ Low (5-20%), ■ Idle (<5%)
- Configurable period: 7, 30, or 90 days analysis window
- Auto-recommendation for lab budget planning (UGC/AICTE)

Visual utilization bars show per-device usage with heartbeat counts.

11.1 Multi-Organization View

For System Integrators managing multiple college/institute clients:

- ■ Navigate to Multi-Org View for cross-org fleet overview
- ■ Each org card shows: device count, online/offline, health score, grades
- ■ Click any org to drill down into per-device details
- ■ Health score = weighted average: A=100, B=80, C=60, D=40, F=20

12. Alert & Notification Config

Configure per-organization notification channels:

- ■ Email — Set SMTP env vars (SMTP_HOST, SMTP_USER, SMTP_PASS)
- ■ Webhook — Slack, Microsoft Teams, or custom webhook URL
- ■ WhatsApp — Requires WhatsApp Business API key (WHATSAPP_API_KEY)

Alert Triggers:

- ■ Tamper/Theft — Hardware component change detected
- ■ Device Offline — No heartbeat received
- ■ Grade Drop — Health grade degradation
- ■ AMC Expiry — Warranty/contract approaching end date

13.1 Network Range Scanner

This is a gated premium feature, locked by default per organization.

To enable:

- Go to Organizations → Select org → Enable 'network-scan' feature flag
- Set access period: 60 days, 365 days, or custom duration

Scanner capabilities (when unlocked):

- ICMP Ping sweep across IP range
- TCP port scanning on discovered hosts
- OS fingerprinting via TTL analysis
- WMI-based remote inventory (Windows hosts, requires credentials)

14. Admin Credentials & Security

Default Login Credentials:

Role: Super Admin

Username: admin

Password: CatalystAdmin@2026

Role: Admin

Username: bostontech

Password: Boston@2026

■ SECURITY RECOMMENDATIONS

1. Change default passwords via ADMIN_PASS / ADMIN_PASS_2 env vars
2. Use HTTPS (already configured via nginx + Let's Encrypt)
3. Restrict dashboard access via IP allowlist in nginx.conf
4. Rotate admin passwords quarterly
5. Monitor /api/auth/login for brute-force attempts

Q: Scanner .exe shows 'License locked'

A: Add scan credits via Licenses page or Bulk License Manager.

Q: Device shows 'Offline' even though PC is running

A: Check network — scanner needs internet to send heartbeats to server.

Q: Dashboard not loading

A: Check if Docker containers are running: `sudo docker compose ps`

Q: Network scan button not visible

A: Feature is locked. Enable via Organizations → Feature Flags.

Q: E-Waste report shows all devices as 'Keep'

A: BIOS date may not be available. Devices need a full scan first.

Q: AICTE CSV download is empty

A: Ensure devices have completed at least one scan with telemetry.

Q: Webhook test fails

A: Verify webhook URL. For Slack, use Incoming Webhooks integration.

Q: Email alerts not sent

A: Configure SMTP env vars in `docker-compose.yml` or `.env` file.

Support Contact

For technical support, contact:

- Email: support@bostontechindia.in
- Dashboard: <https://catalystscan.bostontechindia.in>
- Documentation: Refer to this guide or contact your account manager